Risk across tails and Haar horizons: HiTec + Manuf

Evidence draft for human/team review; NOT submission approved. Question: does fitted tail dependence change with horizon, and how much risk changes when dependence is held invariant or Gaussian? A fixed 2006-2015 model is evaluated on untouched 2016-2025 returns. This is a risk-model comparison, not a trading strategy or causal study.

Data: first value-weighted daily section of Kenneth French 5 Industry Portfolios; sectors HiTec and Manuf. Selected coverage 2006-01-03 through 2025-12-31: 5031 observed dates; training 2517, evaluation 2514. Percentage returns are divided by 100. Missing sentinels -99.99/-999, nonfinite/invalid returns, duplicate and unsorted dates are rejected, not imputed.

Audit: 0 missing values, 0 nonfinite values, 0 duplicate dates. There are 186 absent weekdays between observed endpoints; these include exchange holidays and are not established missing trading sessions. No independent exchange-calendar completeness claim is made. Full year counts and extrema dates are in results.json.

Sector	Minimum daily (%)	Maximum daily (%)	Mean daily (%)
HiTec	-12.560	12.850	0.0646
Manuf	-11.730	14.400	0.0417

Choose two broad US sector research portfolios to keep estimation interpretable: technology and manufacturing provide distinct labels but are strongly exposed to common equity shocks. These are research portfolios, not frictionless tradeable ETFs. Source construction assigns NYSE/AMEX/NASDAQ stocks by SIC at June-end; the present archive is revised historical data, not point-in-time vintages. The library documents its 2025 FIZ-to-CIZ format change.

Causal Haar construction: for h = 1,2,4,8,16, $x_t = \log(1+r_t)$. On each completed disjoint block, orthonormal Haar coarse coefficient $a_h = \text{sum}(x_t)/\sqrt{h}$. Recover the actual sector holding-period return $R_h = \exp(\sqrt{h} \cdot a_h) - 1$. PyWavelets computes the coarsest scaling coefficient within each block. At h=1 the transform is identity. This is a wavelet multiresolution scaling/coarse approximation, NOT a band-pass detail-return study; Haar details describe signed oscillations and cannot be compounded as investable returns.

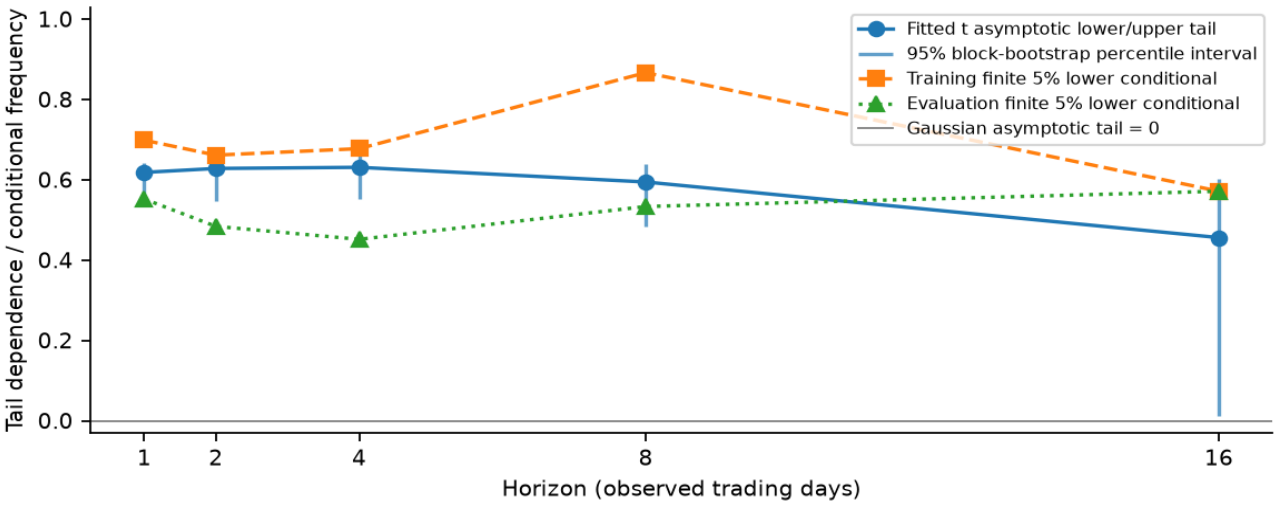
Portfolio starts every h-day block with 50% wealth in each sector and holds those sector exposures until block-end, then rebalances: $R_{p,h} = (R_{HiTec,h} + R_{Manuf,h})/2$; loss $L_h = -R_{p,h}$. This is not daily rebalancing inside multi-day blocks. No annualization, square-root-of-time risk scaling, risk-free subtraction, fees or leverage. Split before transformations; align separately at first training and evaluation observation. Incomplete terminal blocks are discarded. No block straddles the split and no full-sample transform/filter is used.

Dependence estimation and tail uncertainty

Ranks U_ij = rank(R_ij)/(n+1), with average ties, train each horizon independently. Maximize bivariate Student-t copula pseudo-log-likelihood sum(log f_t,2(z1,z2;rho,nu) - log f_t(z1;nu) - log f_t(z2;nu)), z_j=t_nu inverse(U_j), rho bounded [-0.98,0.98], nu grid [2.1, 3, 4, 5, 7, 10, 15, 30, 60]. Gaussian rho is separately fitted by the corresponding normal copula likelihood. Empirical margins stay the same across all simulated copulas. Grid/bounds are numerical/model restrictions, not evidence of a true tail law; boundary fits especially require caution.

Student-t symmetric asymptotic lower/upper tail lambda = 2*t_(nu+1)(-sqrt((nu+1)*(1-rho)/(1+rho))). Gaussian lambda=0 for |rho|<1, but its finite-threshold co-exceedance need not be zero. Finite lower diagnostic is Pr(U_Manuf <= q | U_HiTec <= q), upper analog at 1-q. Each period is ranked independently for diagnostics only; evaluation ranks never set forecasts. Results include q=0.05 and q=0.10 and both tails. Symmetric t cannot represent asymmetric crash/rally dependence.

h	n train	rho t	nu	lambda t	95% lambda interval	n effective*
1	2517	0.821	2.1	0.618	[0.566, 0.642]	473.6
2	1258	0.830	2.1	0.628	[0.547, 0.662]	339.8
4	629	0.833	2.1	0.631	[0.552, 0.670]	362.2
8	314	0.797	2.1	0.594	[0.485, 0.640]	227.3
16	157	0.809	5	0.456	[0.013, 0.603]	157.0



Uncertainty: 199 seeded circular moving-block resamples of paired horizon observations; block length ceil(20/h), approximately one trading month. Refit ranks and t likelihood per training resample, report 2.5/97.5 percentile intervals. Separate evaluation finite-tail resamples never affect forecasts. *Effective n is an initial-positive-autocorrelation heuristic for the lower joint 5% indicator, not a proven count of independent observations. Overlapping intervals alone are not a formal equality test. Horizon estimates are dependent; no formal multiple-horizon significance claim is made.

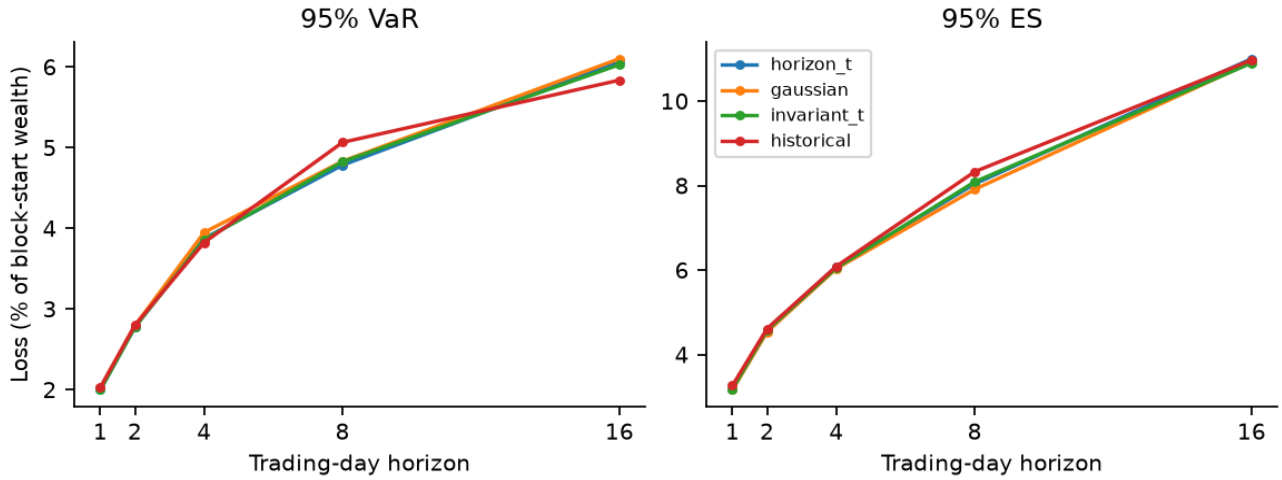
Point estimates vary with horizon, but this sample does not establish a reliable monotone horizon law. Long-horizon intervals are broad and only a few joint-tail events remain. A single stationary fit blends crisis/non-crisis states; uncertainty conditional on this choice does not cover all regime or grid misspecification.

Measured holding-period risk and controlled ablation

Simulate 100,000 paired uniforms per copula with seed 20261005+h, then empirical inverse-CDF sector margins (no extrapolation past training extremes). Compute the equal-weight portfolio loss. VaR is the empirical 95% inverse-CDF loss quantile. ES is the integrated worst 5% loss mass, with a fractional order statistic at the boundary to handle ties exactly. Three deterministic independent Monte Carlo seed batches quantify numerical sensitivity; JSON stores their min/max ranges, not statistical confidence intervals.

Models: horizon_t refits both rho and nu per scale; gaussian removes asymptotic tail dependence but has its own fitted correlation; invariant_t reuses DAILY TRAINING t rho and nu at every horizon with that horizon's SAME empirical margins; historical samples the observed training joint portfolio distribution. The invariant_t comparison isolates scale dependence within the t family. Gaussian comparison changes copula family/shape and cannot alone isolate horizon invariance. Models are frozen after training, with no volatility updating.

h	Model	VaR95 (%)	ES95 (%)
1	horizon_t	2.0000	3.1897
1	gaussian	2.0200	3.1678
1	invariant_t	2.0000	3.1897
1	historical	2.0300	3.2724
2	horizon_t	2.7825	4.5917
2	gaussian	2.7955	4.5413
2	invariant_t	2.7788	4.5809
2	historical	2.8023	4.6191
4	horizon_t	3.8770	6.0785
4	gaussian	3.9513	6.0389
4	invariant_t	3.8620	6.0565
4	historical	3.8263	6.0937
8	horizon_t	4.7860	8.0452
8	gaussian	4.8346	7.9174
8	invariant_t	4.8281	8.0954
8	historical	5.0682	8.3349
16	horizon_t	6.0705	11.0094
16	gaussian	6.1047	10.9275
16	invariant_t	6.0307	10.9086
16	historical	5.8390	10.9774



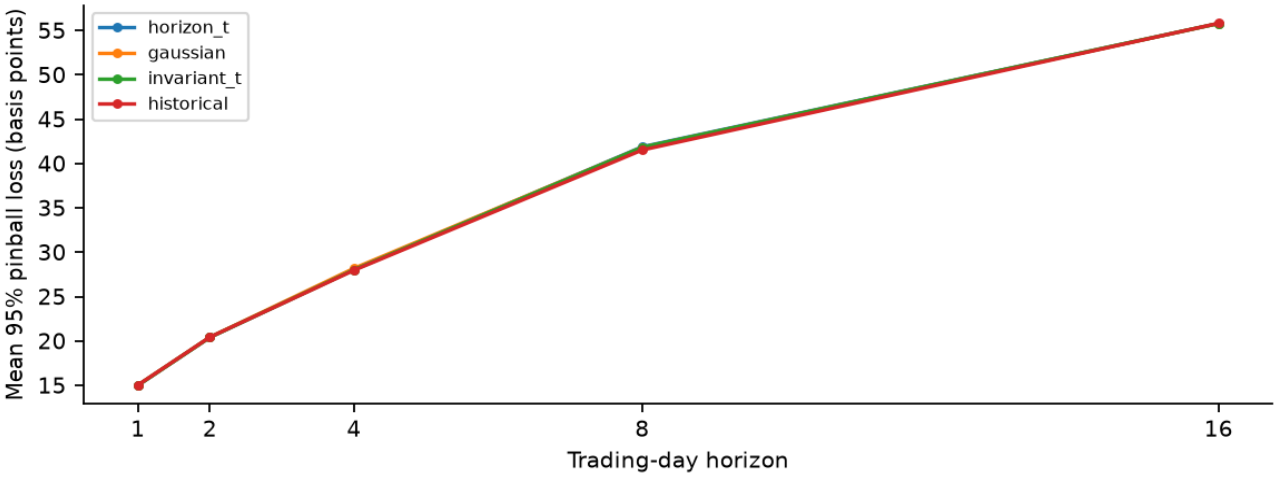
h	t-Gauss VaR (bp)	t-Gauss ES (bp)	t-invariant VaR (bp)	t-invariant ES (bp)
1	-2.00	2.19	0.00	0.00
2	-1.30	5.04	0.37	1.08
4	-7.44	3.95	1.50	2.20
8	-4.86	12.78	-4.21	-5.01
16	-3.43	8.19	3.98	10.08

Positive differences mean the horizon-t risk estimate is larger; signs are not presumed. Tail dependence does not imply every 95% quantile rises: dependence shape, high common correlation, and finite empirical margins matter. Small differences can be comparable to Monte Carlo variation or much smaller than estimation uncertainty. Daily t and invariant-t agree exactly by design.

Actual out-of-sample backtest

Each completed evaluation block contributes one actual portfolio loss; no overlaps within a horizon. Forecasts are constant within each horizon. Breach = observed loss strictly above VaR; target 5%. Exact two-sided binomial p-values and Clopper-Pearson intervals assume independent identically distributed breaches, an assumption not assured by financial time series. Serially dependent data motivate additional moving-block breach-rate intervals in JSON. Multiple models/horizons are descriptive comparisons, not corrected hypothesis tests.

h	Model	n test	Breaches	Rate (%)	Binomial p*	Pinball (bp)
1	horizon_t	2514	100	3.98	0.0173	14.997
1	gaussian	2514	97	3.86	0.0079	15.018
1	invariant_t	2514	100	3.98	0.0173	14.997
1	historical	2514	95	3.78	0.0039	15.030
2	horizon_t	1257	50	3.98	0.1052	20.414
2	gaussian	1257	49	3.90	0.0801	20.428
2	invariant_t	1257	51	4.06	0.1363	20.411
2	historical	1257	48	3.82	0.0601	20.436
4	horizon_t	628	20	3.18	0.0349	28.058
4	gaussian	628	19	3.03	0.0216	28.195
4	invariant_t	628	21	3.34	0.0546	28.032
4	historical	628	21	3.34	0.0546	27.973
8	horizon_t	314	20	6.37	0.2445	41.889
8	gaussian	314	20	6.37	0.2445	41.822
8	invariant_t	314	20	6.37	0.2445	41.831
8	historical	314	19	6.05	0.3646	41.529
16	horizon_t	157	8	5.10	0.8550	55.793
16	gaussian	157	8	5.10	0.8550	55.790
16	invariant_t	157	8	5.10	0.8550	55.797
16	historical	157	8	5.10	0.8550	55.815



95% pinball loss = mean(max(0.95*(L-VaR), -0.05*(L-VaR))); lower is better, in loss units. It jointly penalizes breaches and unnecessarily high quantiles; breach rates alone do not rank models. Different horizon losses have different units of holding-period exposure, so do not interpret increasing cross-horizon pinball as declining model quality.

The horizon-specific t model is not uniformly best against historical or invariant-t. Short-horizon fixed forecasts can be overly conservative, while crisis losses still cluster; a nonsignificant coverage test at a sparse horizon is not proof of calibration. No trading-profit, investable-strategy, causal, or universal superiority claim is supported.

Small tails and ES: diagnostics, not validation

h	Train lower joint / cond.	Train lower freq [95% CI]	Eval lower joint / cond.	Eval lower freq [95% CI]
1	88/126	0.698 [0.587, 0.752]	69/125	0.552 [0.451, 0.672]
2	41/62	0.661 [0.500, 0.794]	30/62	0.484 [0.397, 0.645]
4	21/31	0.677 [0.515, 0.812]	14/31	0.452 [0.290, 0.600]
8	13/15	0.867 [0.498, 0.933]	8/15	0.533 [0.267, 0.734]
16	4/7	0.571 [0.142, 1.000]	4/7	0.571 [0.143, 0.858]

Finite 5% conditional frequencies are not asymptotic lambda estimates. At the longest horizon only a handful of conditioning observations remain, and bootstrap ties may reduce rank counts further. Full upper-tail and 10% diagnostics, effective sample sizes, and incomplete-block counts are retained in results.json.

ES is not validated by the VaR coverage test. Descriptive ES residual = ES - VaR - mean(max(L-VaR,0))/0.05; zero is a calibration moment when VaR and ES are jointly correct (with quantile/tie qualifications). It is not a standalone ES test. Observed mean loss above forecast VaR is also stored, but differs from an unconditional 5%-tail ES when coverage differs. Positive residual means the forecast excess-tail budget is above observed average excess loss, not proven conservatism under future stress.

h	Model	ES residual (%)	95% block interval (%)
1	horizon_t	0.2561	[-0.407, 0.706]
1	gaussian	0.2299	[-0.430, 0.677]
1	invariant_t	0.2561	[-0.407, 0.706]
1	historical	0.3321	[-0.325, 0.778]
2	horizon_t	0.6380	[-0.110, 1.221]
2	gaussian	0.5849	[-0.161, 1.165]
2	invariant_t	0.6280	[-0.121, 1.212]
2	historical	0.6611	[-0.083, 1.240]
4	horizon_t	0.7246	[-0.496, 1.627]
4	gaussian	0.6577	[-0.550, 1.546]
4	invariant_t	0.7078	[-0.515, 1.615]
4	historical	0.7567	[-0.471, 1.676]
8	horizon_t	0.1845	[-2.388, 2.010]
8	gaussian	0.0699	[-2.487, 1.875]
8	invariant_t	0.2461	[-2.313, 2.053]
8	historical	0.5460	[-1.932, 2.255]
16	horizon_t	0.8774	[-4.860, 4.037]
16	gaussian	0.7962	[-4.934, 3.938]
16	invariant_t	0.7758	[-4.971, 3.956]
16	historical	0.8410	[-4.950, 4.120]

h	Pinball t-HS (bp)	Paired 95% block interval (bp)	Pinball t-invariant (bp)	Paired interval (bp)
1	-0.0331	[-0.064, 0.005]	0.0000	[0.000, 0.000]
2	-0.0216	[-0.046, 0.007]	0.0037	[-0.001, 0.008]
4	0.0848	[-0.003, 0.165]	0.0257	[0.001, 0.050]
8	0.3595	[-0.243, 1.241]	0.0577	[-0.036, 0.192]
16	-0.0221	[-0.907, 0.715]	-0.0038	[-0.156, 0.123]

Paired bootstrap intervals resample evaluation loss differences at common block indices, conditional on frozen forecasts; they do not include training-parameter uncertainty or correct for multiple comparisons. Bootstrap length is a transparent fixed choice rather than optimally estimated; sensitivity to other lengths/regimes remains future work. Historical empirical margins truncate possible novel crashes and simulation cannot fix that.

Recommendation, provenance and reproducibility

Answer: fitted symmetric t tail dependence changes from 0.618 at 1 day to 0.456 at 16 days, with longest-horizon 95% interval [0.013, 0.603]. This is suggestive variation, not established horizon decay: there are only 157 training blocks and 4 joint 5% lower-tail events at that horizon. Gaussian and invariant-t risk differences are measured separately above; ignoring tails is not shown to mechanically understate every 95% VaR.

Actionable recommendation: retain historical simulation as a transparent baseline and add an explicit horizon-specific t-versus-daily-t/ Gaussian ES sensitivity panel to each holding-period risk review. Do not reduce long-horizon limits because fitted lambda fell; sparse tails and wide intervals do not support that decision. Require a separate rolling/volatility-updated, point-in-time validation before production use, rather than declaring this frozen model superior.

Main limitations: one two-sector portfolio, one split, high dependence, coarse df grid (some lower-bound fits), symmetric tails, rank ties, finite empirical support, no volatility/regime updating, overlapping information across scales, month-length bootstrap choice, no parameter-uncertainty interval for VaR/ES, and revised historical source vintages. Strict chronological fitting prevents algorithmic look-ahead, not vintage revision bias. Sensitivity ranges across three Monte Carlo batches measure only numerical randomness.

One command after installing locked dependencies: python -m src.study. It acquires/pin-checks data and regenerates results JSON/CSV, all three figures, this markdown/PDF evidence report and provenance. python -m src.study --offline uses the validated local cache; --refresh checks upstream and fails clearly on revision without replacing the pinned cache. No raw or selected French data is distributed. A fresh download after upstream changes may be blocked until the snapshot is deliberately reviewed and repinned; an existing validated local cache is the fallback, not a promise of perpetual public-byte availability.

Source: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/5_Industry_Portfolios_daily_CSV.zip

Accessed UTC: 2026-10-05T11:17:16.255084+00:00; source archive header: This file was created using the 202608 CRSP database.

Archive SHA256: 8ede64c6df9134ab5c372a264f31640ab6737b9f942f7207f62828005542c9d3

Canonical selected snapshot SHA256: 4ac1f03a05e875801eecc56329cb52ed16e3b89b1277a8004b4791d4d651c48a

Selected snapshot serialization: date,HiTec,Manuf CSV; ISO dates, decimal returns at 8 fractional digits, LF line endings. Source detail/library pages carry Fama/French copyright; public accessibility is NOT proof of an open redistribution licence. Consult source terms independently. Public download code plus aggregate analytical outputs are provided, not French raw tables.

References: Kenneth French Data Library, 5 Industry Portfolios detail: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_5_ind_port.html ; library revision/format notes: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html ; PyWavelets Haar multiresolution implementation; SciPy rankdata, t distributions, bounded optimization and exact binomtest. All statistical formulas used are stated here and in code.

AI disclosure: Hermes Agent (Nous Research), OpenAI gpt-6.1-sol, materially assisted methodology, implementation, tests, execution, figures and report drafting. Tests and real downloaded data were executed; independent human/team review and submission approval remain outstanding. Team members must understand and be able to modify and reproduce all calculations. See docs/AI_USAGE.md and evidence/ for red/green and reproduction logs.